

Nanopores

How to cite: *Angew. Chem. Int. Ed.* **2022**, 61, e202209970

International Edition: doi.org/10.1002/anie.202209970

German Edition: doi.org/10.1002/ange.202209970

3D Blockage Mapping for Identifying Familial Point Mutations in Single Amyloid- β Peptides with a Nanopore

Kai-Li Xin, Zheng-Li Hu,* Shao-Chuang Liu, Xin-Yi Li, Jun-Ge Li, Hongyan Niu, Yi-Lun Ying, and Yi-Tao Long*

Abstract: Accurate discrimination of amyloid- β (A β) peptides containing familial point mutations would advance the knowledge of their roles in early-onset Alzheimer's disease. Herein, we simultaneously identified the mutant A21G, E22G, E22Q, and the wild-type (WT) A β_{18-26} peptides with aerolysin nanopore using a 3D blockage mapping strategy. The standard deviation of current blockade fluctuations (σ_b) was proposed as a new supplement to current blockage (I_b/I_0) and duration time (t_D) to profile the blockage characteristics of single molecules. Although the WT and A21G A β_{18-26} are indistinguishable in a traditional I_b/I_0 - t_D 2D description, ~87% of the blockade events can be accurately classified with half reduction of false identification using a combination of I_b/I_0 , t_D , and σ_b . This work offers an easy and reliable strategy to promote nanopore sensitivity of peptide mutants, leading to a more precise analysis of pathogenic mutations for developing effective diagnosis and treatment.

Introduction

Familial Alzheimer's disease is associated with single amino acid mutations in the amyloid- β (A β) peptides, which could alter the neurotoxicity and increase the risk to exacerbate disease development.^[1] The familial A β mutations are varied in individual patients and may contribute to unique clinical phenotypes, such as the Flemish (A21G),^[2] Arctic (E22G),^[3] and Dutch (E22Q) versions.^[4] To date, the characterization of A β variants is still difficult because the targets of interest

usually show low abundance and small differences in molecular masses and charges.^[5] Nanopore-based single-molecule techniques have been developed as promising alternatives for the precise measurements of various biomolecules including proteins and peptides at high sensitivity without the requirements of complex operations and expensive equipment.^[6] Biological nanopores, such as α -hemolysin,^[7] aerolysin,^[8] Fragaceatoxin C (FraC),^[9] and Outer Membrane Porin F (OmpF) nanopores,^[10] have been used to discriminate peptides with different lengths. Some of them were also applied to study peptide/protein post-translational modifications including phosphorylation,^[11] glycosylation,^[12] acetylation,^[13] and methylation.^[14] Recently, the aerolysin nanopore successfully identified the variations of single standard amino acids, which were purposefully positioned at the C-terminus of a polyarginine carrier.^[15] By employing a DNA helicase or polymerase motor to control the translocations of the specially designed DNA-peptide conjugates, several proof-of-concept studies demonstrated the capability of *Mycobacterium smegmatis* porin A (MspA) nanopores in single-molecule peptide and protein sequencing.^[16] In previous works, nanopores were mainly adopted to probe the model peptides based on differences in molecular lengths or charges. To address the challenge of insufficient sensitivity of nanopores for similar peptides, chemical labels were introduced to specific amino acids (e.g., cysteine) to produce readily distinguishable current responses.^[17] However, to our best knowledge, few works have reported label-free detection of familial disease-linked peptides and variants with the same length and identical net charge by using nanopores.

Nanopore sensing is primarily based on analyzing the regulation of ion flows induced by single molecules traversing through a nanopore driven by a bias voltage. Generally, the ionic current signals are described by two main parameters, namely the current blockade amplitude and duration time, to unravel single analyte's characteristics including chemical compositions and molecular structures.^[18] Numerous efforts have been devoted to amplifying the current blockage and/or duration time differences between different analytes, including modifications of the nanopore sensing interface^[19] and manipulations of the ionic strength or pH of the electrolyte solution.^[20] Besides the current blockade amplitude and duration time, the nanopore current "noise" has become an attractive subject of recent investigations, which submerges rich information on analyte-nanopore-ion interactions.^[21] The nanopore current "noise" is found to be analyte-dependent probably due to different

[*] K.-L. Xin, Dr. Z.-L. Hu, Dr. S.-C. Liu, X.-Y. Li, J.-G. Li, H. Niu, Prof. Y.-L. Ying, Prof. Y.-T. Long
 State Key Laboratory of Analytical Chemistry for Life Science,
 School of Chemistry and Chemical Engineering,
 Nanjing University
 Nanjing 210023 (P. R. China)
 E-mail: zhengliliu@nju.edu.cn
 yitaolong@nju.edu.cn

Dr. Z.-L. Hu
 Shenzhen Research Institute of Nanjing University
 Shenzhen 518057 (P. R. China)

Prof. Y.-L. Ying
 Chemistry and Biomedicine Innovation Center,
 Nanjing University
 Nanjing 210023 (P. R. China)

interaction dynamics occurring within the nanopore sensing space,^[22] which can lead to varied fluctuation degrees of the ionic current blockades (Figure 1). To discover more dimensional information for the enhanced discrimination of the A β peptides bearing familial mutations, herein, we established a 3D data mapping method for describing the current blockade features, where the standard deviation of ionic current blockade fluctuations (σ_b) was introduced as a third parameter for the accurate and efficient nanopore analysis. This strategy provides a feasible way to profile and classify individual signals with undetectable differences in the current blockade amplitudes, bringing fresh insights into nanopore sensing of single molecules with very subtle variations in masses and charges.

Results and Discussion

As shown in Figure 1c, the σ_b values of individual current blockade events are calculated by the following Equation (1),

$$\sigma_b = \sqrt{\frac{1}{N} \sum_{i=1}^N (I_i - \bar{I})^2} \quad (1)$$

where N is the number of sampling points, I_i is the residual current at sampling point i , and $\bar{I}$ is the mean value of the residual current. To verify the feasibility and reliability of σ_b for nanopore sensing of single molecules, the Azo-ODN was exploited as the model, which contains a photoisomerizable azobenzene group covalently linked to the deoxyribose in the middle position, and a poly(dA)₂ segment in each terminus, respectively (Figure 2a, top). When added to the *cis* chamber, Azo-ODN molecules were captured into the wild-type aerolysin nanopore, producing characteristic current signals with relatively uniform blockade amplitudes because the majority of Azo-ODN were observed in the

trans form (Figure S3). After irradiation with UV light ($\lambda = 365$ nm), typical blockade events with a reduced current blockage were easily discovered for the *cis*-Azo-ODN together with those for the *trans* isomers (Figure 2a, middle, and Figure S3). Along with the continuous translocations of the Azo-ODN photoisomers that owe the same molecular mass, ion distributions inside the nanopore confined space rearranged accordingly, causing visible differences in ionic current blockade fluctuations for the *trans* and *cis* conformers (Figure 2a, bottom). Here, the normalized current blockage is defined as I_b/I_0 , where I_b represents the ionic current blockade amplitude, I_0 indicates the open pore current, and t_D is the duration time of single-molecule translocation, respectively. With a combination of I_b/I_0 , $\log(t_D)$, and σ_b features, the 3D scatter plot illustrates two distinguishable populations of typical blockade events belonging to the *trans*-Azo-ODN and the *cis*-Azo-ODN (Figure 2b). Although the duration time manifested remarkably overlap of nearly 100 % (Figure S4b), the excellent separations of I_b/I_0 and σ_b (Figure S4a and S4c) enabled direct discrimination of the *trans* and *cis* isomers of Azo-ODN in the 2D density plots of I_b/I_0 vs. $\log(t_D)$ (Figure 2c) and $\log(t_D)$ vs. σ_b (Figure 2d). As expected, two well-defined populations were found in the 2D density plot of I_b/I_0 vs. σ_b (Figure 2e), further supporting that the newly proposed parameter of σ_b is practical and potential for directly discriminating similar translocating molecules.

Then, we utilized σ_b to identify the WT A β_{18-26} and its A21G mutant, which have the same net charge of -2.78 at pH 8.0 and only differ by a single methyl group (Table S1 and Figure 3a). The two kinds of peptides produced extremely short duration time ($t_D < 0.3$ ms at $+120$ mV) when detected by the wild-type aerolysin nanopore (Figure S5), which is impossible to obtain reliable σ_b values for further analysis of A β_{18-26} peptides containing single amino acid mutation. The aerolysin nanopore needs to be engineered for a strong sensing ability. The mutant T232K/K238Q aerolysin nanopore was revealed to slow down the

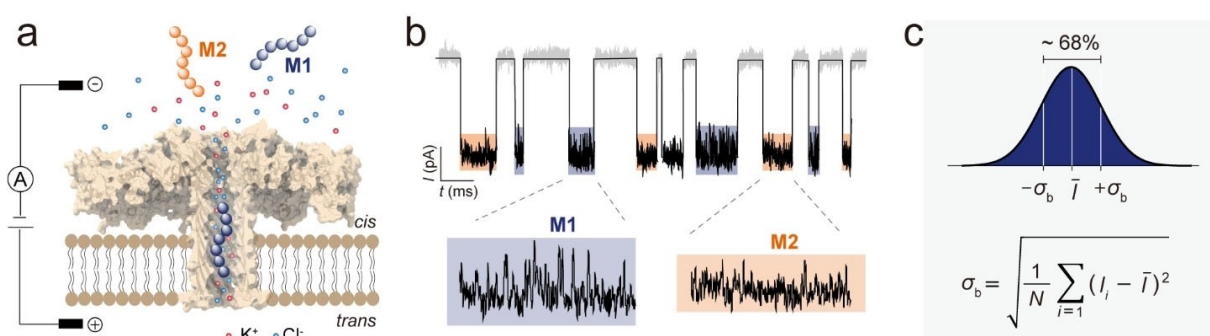


Figure 1. Single-molecule sensing with aerolysin nanopore based on the standard deviation of current blockade fluctuations (σ_b). a) Single-channel recording of Molecule 1 (M1) and Molecule 2 (M2) using an aerolysin nanopore. M1 and M2 were forced to enter and translocate the nanopore from the *cis* solution by the voltages applied via a pair of Ag/AgCl electrodes. b) The analytes produced signals with different current blockade characteristics, and the zoomed segments of two typical M1 and M2 signals show distinct current blockade fluctuations. c) Calculation of σ_b for individual current blockade events. As the residual currents collected from every sampling point of a blockade signal are usually fitted into a normal distribution, σ_b was used to present the variations of the current blockade segment, where $\sim 68\%$ of the recorded values are placed within one standard deviation from the mean residual current. The calculated equation for σ_b is shown at the bottom, where N , I_i and $\bar{I}$ represent the number of sampling points, the residual current at sampling point i , and the average value of the residual currents, respectively.

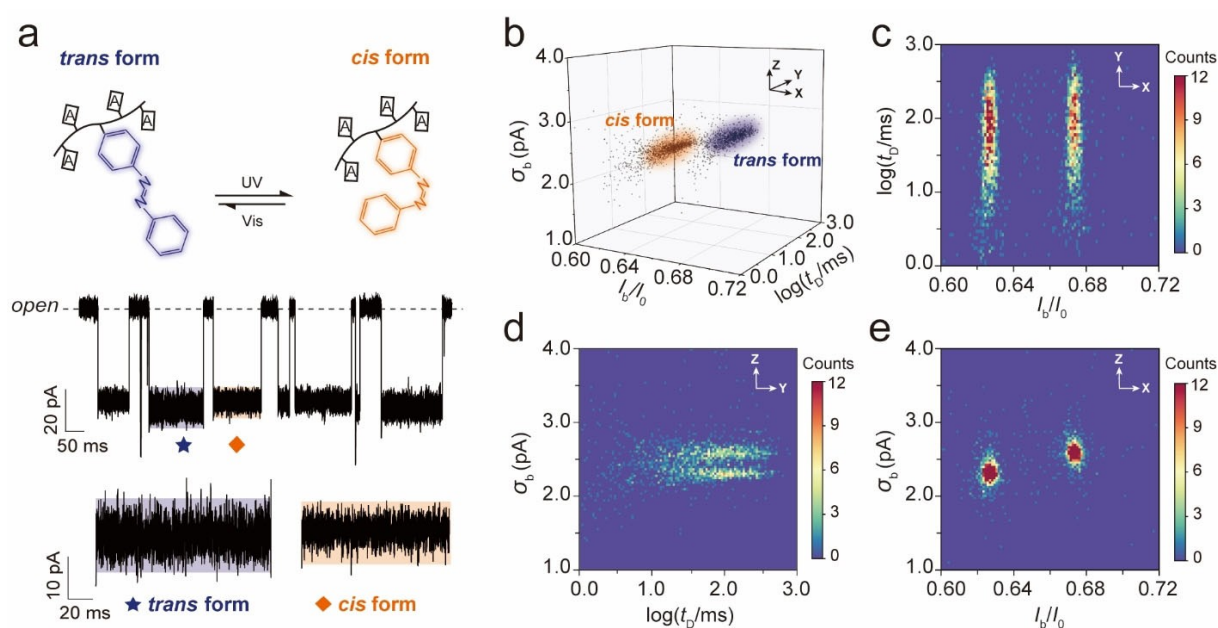


Figure 2. Discrimination of Azo-ODN isomers by a wild-type aerolysin nanopore. a) Top: Illustration of the *trans/cis* conversion of Azo-ODN regulated by UV/Vis light. Middle: Current traces of single Azo-ODN translocations. Bottom: The zoomed-in current blockade segments of the *trans*- (blue bands) and *cis*-Azo-ODN (orange bands) typical events. b) 3D scatter plot based on I_b/I_0 , $\log(t_b)$ and σ_b of Azo-ODN current blockade events. The three features can assign Azo-ODN blockades into two distinct *trans* and *cis* groups. Each point in the scatter plot represents a typical blockade event. Event density plots of I_b/I_0 vs. $\log(t_b)$ (c), $\log(t_b)$ vs. σ_b (d), and I_b/I_0 vs. σ_b (e) for Azo-ODN isomers. The inserted color code reflects the local density of current blockade events. The final concentration of Azo-ODN was fixed at 2.0 μM . Data were obtained at +120 mV in the buffer containing 1 M KCl, 1 mM EDTA, and 10 mM Tris, pH 8.0.

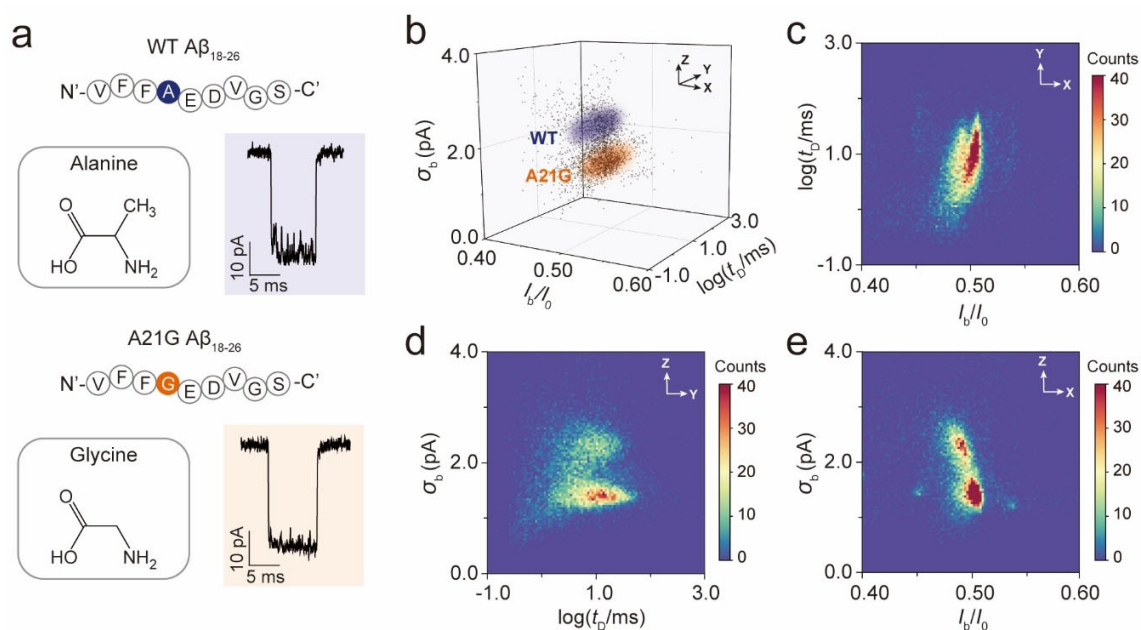


Figure 3. Detection of the WT and A21G $\text{A}\beta_{18-26}$ peptides by the engineered T232K/K238Q aerolysin nanopore. a) Representative current blockade events of the WT and the A21G $\text{A}\beta_{18-26}$. The sequences are $\text{N}'\text{-VFFAEDVGS-C}'$ for the WT $\text{A}\beta_{18-26}$, and $\text{N}'\text{-VFFGEDVGS-C}'$ for the A21G $\text{A}\beta_{18-26}$, respectively. Lower left: Molecular structures of alanine and glycine. b) 3D scatter plot of I_b/I_0 vs. $\log(t_b)$ vs. σ_b for the WT and A21G $\text{A}\beta_{18-26}$ peptides. Event density plots of I_b/I_0 vs. $\log(t_b)$ (c), $\log(t_b)$ vs. σ_b (d), and I_b/I_0 vs. σ_b (e) for the WT and A21G $\text{A}\beta_{18-26}$ peptides. Each peptide was added into the *cis* chamber at a final concentration of 10.0 μM . Data for $\text{A}\beta_{18-26}$ peptides were acquired at +120 mV in the buffer containing 1 M KCl and 10 mM Tris, pH 8.0.

translocation speeds by ~ 50.3 folds for the WT $A\beta_{18-26}$ peptide and ~ 39.6 folds for the A21G $A\beta_{18-26}$ peptide at $+120$ mV in comparison to the wild-type aerolysin nanopore, providing sufficient long duration time to accurately extract σ_b features of single-molecule signals. Our previous work has demonstrated that the mutant T232K/K238Q aerolysin nanopore is sensitive to the negatively charged peptides due to the reduction in translocation speeds.^[11a] In addition, the voltage-dependent studies unraveled that $+120$ mV as one optimal potential contributed to a relatively larger σ_b difference of single peptide signals with long enough duration time for discriminating the WT and A21G $A\beta_{18-26}$ peptides by this mutant aerolysin nanopore (Figure S6). Therefore, we employed the engineered T232K/K238Q aerolysin nanopore for the subsequent measurements of the WT and the mutant $A\beta_{18-26}$ peptides at $+120$ mV.

When the WT and the A21G $A\beta_{18-26}$ peptides were captured into the T232K/K238Q aerolysin nanopore, single-molecule signals were observed to exhibit very similar current blockade amplitudes and duration time (Figure 3a and Figure S7). In the 3D scatter plot that was depicted by the parameters of σ_b together with I_b/I_0 and $\log(t_D)$, the two separated populations were observed, which indicates that the WT and the A21G $A\beta_{18-26}$ could be discriminated from each other (Figure 3b). However, in the conventional 2D density plot of I_b/I_0 vs. $\log(t_D)$, the current blockade events were indistinguishable due to the largely overlapped distributions for the two kinds of $A\beta_{18-26}$ peptides (Figure 3c, Figure S8a and S8b). Interestingly, in the 2D density plots

based on the parameter of σ_b , the current blockade events could be clearly classified into two populations (Figure 3d, e, and Figure S8c). Furthermore, we used a support vector machine (SVM) to classify the two kinds of blockade events with distinct parameter combinations (see Supporting Information). The results showed that the identification accuracies are greatly improved with the introduction of σ_b , where 86.9 % and 87.6 % of the blockade events were retrieved accurately in the 2D density plots of $\log(t_D)$ vs. σ_b and I_b/I_0 vs. σ_b , respectively. In contrast, the false identification rate is reduced by nearly half (Table S3), which further indicates the validity of the third parameter in the nanopore discrimination. Together, these results demonstrated that σ_b contributes to an improved sensitivity for mutant $A\beta$ peptide identification.

Further experiments were conducted to examine the ability of σ_b to recognize the WT and the mutant A21G $A\beta_{18-26}$ peptides in the mixtures. The concentration of the A21G $A\beta_{18-26}$ was fixed in the *cis* chamber, and the WT $A\beta_{18-26}$ was continuously added into the same compartment with the molar concentration ratios of [A21G]:[WT] increasing from 1:1, 1:2 to 1:3. As expected, one population was observed in each 2D density plot of I_b/I_0 vs. $\log(t_D)$ (Figure 4a), while two separated populations were found in each 2D density plot of I_b/I_0 vs. σ_b (Figure 4c), which is consistent with the above results of the pure samples. According to the σ_b features of the pure samples, we assigned the Population 1 (P1) to the current blockade events of the A21G $A\beta_{18-26}$ peptides, and the Population 2 (P2) to the current blockade events of the WT $A\beta_{18-26}$

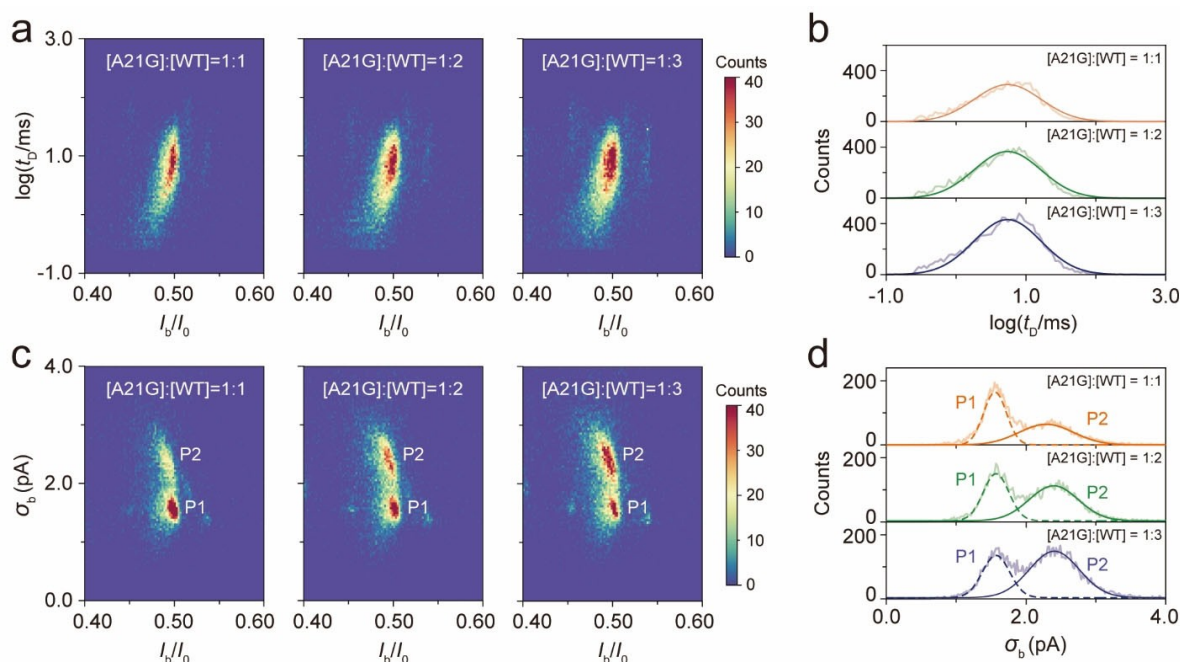


Figure 4. Simultaneous measurement of the WT and A21G $A\beta_{18-26}$ peptides in the mixtures with the mutant T232K/K238Q aerolysin nanopore. Event density plots of I_b/I_0 vs. $\log(t_D)$ (a), and I_b/I_0 vs. σ_b (c) for the WT and A21G $A\beta_{18-26}$ mixtures at varied molar concentration ratios of [A21G]:[WT] = 1:1, 1:2, 1:3, respectively. Histograms of $\log(t_D)$ (b), and σ_b (d) for the $A\beta_{18-26}$ mixtures. Typical blockade events ($N > 7000$) were collected during a recording time of 5 minutes for every density plot or histogram. The final concentration of the mutant A21G $A\beta_{18-26}$ peptide was fixed at $5.0 \mu\text{M}$ in the mixtures. Data were recorded at $+120$ mV in buffer of 1 M KCl and 10 mM Tris, pH 8.0.

peptides, respectively. As a result, when raising the [A21G]:[WT] ratios, one peak composed of the indiscernible P1 and P2 was unraveled with increasing peak areas in the $\log(t_D)$ histograms (Figure 4b), while two distinct peaks of P1 for the A21G A β_{18-26} and P2 for the WT A β_{18-26} were observed with altering relative proportions of peak areas in the σ_b histograms (Figure 4d). The relative proportions of peak areas for P1 ($P_1(\%)$) and P2 ($P_2(\%)$) were calculated from the Gaussian fits of the corresponding σ_b distributions in each histogram (Figure S9). As duration time of the two peptides are almost identical, the capture rates were estimated by Equation (2) for the A21G A β_{18-26} and Equation (3) for the WT A β_{18-26} , as follows:

$$f_{A21G} = \frac{P_1(\%) \times N_{total}}{t} \quad (2)$$

$$f_{WT} = \frac{P_2(\%) \times N_{total}}{t} \quad (3)$$

where N_{total} represents the cumulative number of current blockade events for the WT and the mutant A β_{18-26} peptides, and t is the recording time of current traces. Due to a fixed concentration in the mixtures, the calculations showed similar capture rates of $15.55 \pm 0.64 \text{ s}^{-1}$ for the A21G A β_{18-26} . Whereas, with growth in the [A21G]:[WT] ratios, the capture rates for the WT A β_{18-26} linearly increased from $12.83 \pm 1.27 \text{ s}^{-1}$ to $28.90 \pm 1.15 \text{ s}^{-1}$, which indicates quantification is permitted for the WT and the mutant A β peptides relied on the parameter of σ_b in the mixture samples.

Subsequently, σ_b was expanded to the identification of the mutant A21G, E22Q, E22G, and the WT A β_{18-26} peptides with the aerolysin nanopore (Figure 5a and Figure S10). Analysis revealed the relatively clearest separations for the E22G and the A21G A β_{18-26} peptides in the 1D I_b/I_0 histogram (Figure 5b), the E22Q and the WT A β_{18-26} peptides in the 1D σ_b histogram (Figure 5d), but a poor separation for any two kinds of the A β_{18-26} peptides in the 1D $\log(t_D)$ histogram (Figure 5c). As expected, only the E22G A β_{18-26} peptide could be evidently determined in the 2D I_b/I_0 - $\log(t_D)$ scatter plot (Figure 5e). All the four kinds

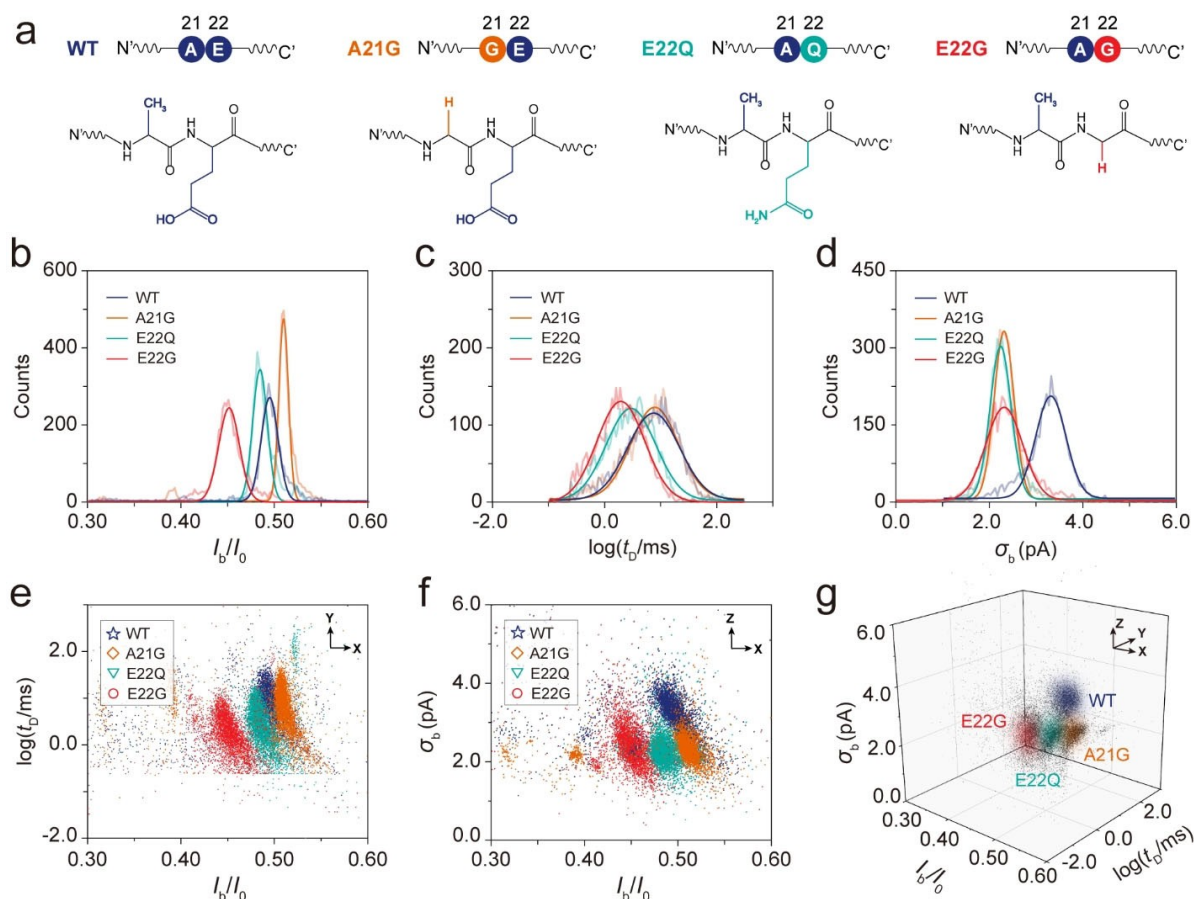


Figure 5. Single-molecule analysis of A β_{18-26} peptides with single amino acid mutations at the A21 or E22 site using the T232K/K238Q aerolysin nanopore. a) Molecular structures of the WT, A21G, E22Q and E22G A β_{18-26} peptides. Histograms of I_b/I_0 (b), $\log(t_D)$ (c) and σ_b (d) for the WT and the mutant A β_{18-26} peptides (from left to right). 2D density plots of I_b/I_0 vs. $\log(t_D)$ (e), and I_b/I_0 vs. σ_b (f) for the four kinds of A β_{18-26} peptides. g) 3D scatter plot of I_b/I_0 vs. $\log(t_D)$ vs. σ_b for the WT A β_{18-26} and the mutant A21G, E22Q and E22G A β_{18-26} peptides. Typical blockade events ($N = 4000$) of every peptide were obtained to draw each 1D histogram, 2D scatter plot and 3D scatter plot. The final concentration was fixed at $5.0 \mu\text{M}$ for each type of the A β_{18-26} peptide. Data were collected at $+120 \text{ mV}$ in electrolyte solution of 1 M KCl and 10 mM Tris , pH 8.0.

of A β_{18-26} peptides can be well differentiated from each other in the 2D I_b/I_0 - σ_b scatter plot (Figure 5f), which is mainly attributed to the complementary role of σ_b for I_b/I_0 in nanopore discrimination of similar biomolecules. Because of the satisfactory separation, direct identification of the mutant A21G, E22Q, E22G and the WT A β_{18-26} peptides in the pure or mixture samples can be realized in the 3D scatter plots with a combination of the I_b/I_0 , $\log(t_b)$ and σ_b features (Figure 5g and Figure S11). Therefore, it could be concluded that σ_b , as an easily available new parameter extracted from individual current blockade events, is beneficial for effectively improving nanopore ability to resolve peptides of the same length with single amino acid differences. The 3D blockage mapping strategy opens a door for an enhancement in nanopore sensitivity of single molecules by extending the descriptions of current blockade signals.

Conclusion

In summary, we developed a 3D data mapping strategy for individual current blockage signals to enhance nanopore sensitivity for identifying A β peptides with familial point mutations. By the introduction of a third parameter σ_b to evaluate ionic current fluctuation degrees of individual blockade events, the direct distinction of the mutant A21G, E22Q, E22G, and the WT A β_{18-26} peptides was achieved with the aerolysin nanopore. In particular, the parameter of σ_b allows an improved identification accuracy for the WT and A21G A β_{18-26} in the mixture, which reduces the false identification rate by half. Previous works have used the standard deviation of blocking currents in the 2D scatter plots for the characterization of multiple analytes including peptides and proteins.^[23] For example, a 2D scatter plot of the event noise (σ_{ib}) from the standard deviation of the ionic currents during the event vs. the excluded current ($I_{ex}\%$) was applied to distinguish and determine the peptide fragments produced by trypsinated lysozyme.^[23c] However, to our best knowledge, this work is the first report of 3D data mapping for peptide identification based on the combination of I_b/I_0 , $\log(t_b)$ and σ_b , where σ_b embodies a unique information dimension that different from the current blockage and duration time. In addition, the A β_{18-26} peptides with a small volume difference (e.g., WT and A21G; only differing by a methyl group) produced very different and discernible σ_b features, while some others (e.g., E22Q and E22G) with a larger volume difference generated almost the same σ_b values. Therefore, our results suggest that the ionic current blockade fluctuations may be regulated by multiple synergistic factors. Our previous works have experimentally and theoretically demonstrated that the ionic current blockades are the cooperative results of factors including volume exclusion effects and nanopore-analyte interactions, which are possibly associated with the molecular dynamic conformational changes.^[24] These changes of individual peptides could alter the distance, orientation, and contacting volume of amino acid residues in the peptide and in the nanopore interface, leading to modulations in the mode and degree of

the characteristic interactions between the nanopore and the peptides. In this aspect, the σ_b difference is likely ascribed to the synergistic results of the dynamic conformational changes of the peptides and the characteristic interactions between nanopore and the peptides. In the future, joint efforts of the nanopore experiments and molecular dynamics simulations would be devoted to uncovering the relationships between the ionic current fluctuations, molecular conformational changes, and nanopore-analyte interactions. In the pursuit of sufficient sensitivity and high accuracy for nanopore analysis, the 3D blockage mapping method would benefit nanopore technology in the label-free and precise measurements of familial degenerative disease-linked mutations in peptides not limited to A β peptides at a single-molecule level, advancing the implementation of primary prevention and effective treatments.

Acknowledgements

Z. L. Hu thanks the supports from the National Natural Science Foundation of China (22106066), and the Guangdong Basic and Applied Basic Research Foundation (2020A151110298). Y. T. Long and S. C. Liu thank the support from the National Natural Science Foundation of China (22027806, 22090054 and 21834001). We would specially thank Xue-Yuan Wu and Jia Wang for their kind help in protein engineering and oligonucleotide detection.

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords: 3D Blockage Mapping · Aerolysin Nanopore · Amyloid- β Peptides · Familial Alzheimer's Disease · Single Amino Acid Mutations

- [1] a) X. Yang, G. Meisl, B. Frohm, E. Thulin, T. P. Knowles, S. Linse, *Proc. Natl. Acad. Sci. USA* **2018**, *115*, E5849–E5858; b) C. Arber, J. Toombs, C. Lovejoy, N. S. Ryan, R. W. Paterson, N. Willumsen, E. Gkanatsiou, E. Portelius, K. Blennow, A. Heslegrave, J. M. Schott, J. Hardy, T. Lashley, N. C. Fox, H. Zetterberg, S. Wray, *Mol. Psychiatry* **2020**, *25*, 2919–2931; c) A. Hatami, S. Monjaze, S. Milton, C. G. Glabe, *J. Biol. Chem.* **2017**, *292*, 3172–3185.
- [2] L. Hendriks, C. M. van Duijn, P. Cras, M. Cruts, W. Van Hul, F. van Harskamp, A. Warren, M. G. McInnis, S. E. Antonarakis, J.-J. Martin, A. Hofman, C. Van Broeckhoven, *Nat. Genet.* **1992**, *1*, 218–221.
- [3] C. Nilsberth, A. Westlind-Danielsson, C. B. Eckman, M. M. Condron, K. Axelman, C. Forsell, C. Sten, J. Luthman, D. B.

- Teplow, S. G. Younkin, J. Näslund, *Nat. Neurosci.* **2001**, *4*, 887–893.
- [4] E. Levy, M. D. Carman, I. J. Fernandez-Madrid, M. D. Power, I. Lieberburg, S. G. van Duinen, G. T. A. Bots, W. Luyendijk, B. Frangione, *Science* **1990**, *248*, 1124–1126.
- [5] a) B. Veerabhadrapa, C. Delaby, C. Hirtz, J. Vialaret, D. Alcolea, A. Lleó, J. Fortea, M. S. Santosh, S. Choubey, S. Lehmann, *Crit. Rev. Clin. Lab. Sci.* **2020**, *57*, 99–113; b) M. L. DeMarco, Q. Nguyen, A. Fok, G. Y. R. Hsiung, J. G. van der Gugten, *Alzheimer's Dementia* **2020**, *12*, e12036.
- [6] a) M. Afshar Bakshloo, J. J. Kasianowicz, M. Pastoriza-Gallego, J. Mathé, R. Daniel, F. Piguet, A. Oukhaled, *J. Am. Chem. Soc.* **2022**, *144*, 2716–2725; b) F. L. R. Lucas, R. C. A. Versloot, L. Yakovlieva, M. T. Walvoort, G. Maglia, *Nat. Commun.* **2021**, *12*, 5795.
- [7] X. Chen, Y. Zhang, X. Guan, *Nanoscale* **2021**, *13*, 13658–13664.
- [8] F. Piguet, H. Ouldali, M. Pastoriza-Gallego, P. Manivet, J. Pelta, A. Oukhaled, *Nat. Commun.* **2018**, *9*, 966.
- [9] G. Huang, A. Voet, G. Maglia, *Nat. Commun.* **2019**, *10*, 835.
- [10] J. Wang, J. A. Bafna, S. P. Bhamidimarri, M. Winterhalter, *Angew. Chem. Int. Ed.* **2019**, *58*, 4737–4741; *Angew. Chem.* **2019**, *131*, 4788–4792.
- [11] a) S. Li, X.-Y. Wu, M.-Y. Li, S.-C. Liu, Y.-L. Ying, Y.-T. Long, *Small Methods* **2020**, *4*, 2000014; b) C. B. Rosen, D. Rodriguez-Larrea, H. Bayley, *Nat. Biotechnol.* **2014**, *32*, 179–181; c) Y.-L. Ying, J. Yang, F.-N. Meng, S. Li, M.-Y. Li, Y.-T. Long, *Research* **2019**, 1050735.
- [12] L. Restrepo-Pérez, C. H. Wong, G. Maglia, C. Dekker, C. Joo, *Nano Lett.* **2019**, *19*, 7957–7964.
- [13] M.-Z. Huo, Z.-L. Hu, Y.-L. Ying, Y.-T. Long, *Proteomics* **2022**, *22*, 2100041.
- [14] T. Ensslen, K. Sarthak, A. Aksimentiev, J. C. Behrends, *J. Am. Chem. Soc.* **2022**, *144*, 16060–16068.
- [15] H. Ouldali, K. Sarthak, T. Ensslen, F. Piguet, P. Manivet, J. Pelta, J. C. Behrends, A. Aksimentiev, A. Oukhaled, *Nat. Biotechnol.* **2020**, *38*, 176–181.
- [16] a) H. Brinkerhoff, A. S. Kang, J. Liu, A. Aksimentiev, C. Dekker, *Science* **2021**, *374*, 1509–1513; b) S. Yan, J. Zhang, Y. Wang, W. Guo, S. Zhang, Y. Liu, J. Cao, Y. Wang, L. Wang, F. Ma, P. Zhang, H.-Y. Chen, S. Huang, *Nano Lett.* **2021**, *21*, 6703–6710; c) Z. Chen, Z. Wang, Y. Xu, X. Zhang, B. Tian, J. Bai, *Chem. Sci.* **2021**, *12*, 15750–15756.
- [17] L. Restrepo-Pérez, G. Huang, P. R. Bohländer, N. Worp, R. Eelkema, G. Maglia, C. Joo, C. Dekker, *ACS Nano* **2019**, *13*, 13668–13676.
- [18] a) M.-Y. Li, Y.-Q. Wang, Y.-L. Ying, Y.-T. Long, *Chem. Sci.* **2019**, *10*, 10400–10404; b) N. S. Galenkamp, A. Biesemans, G. Maglia, *Nat. Chem.* **2020**, *12*, 481–488.
- [19] a) S. Zhang, G. Huang, R. C. A. Versloot, B. M. H. Bruininks, P. C. T. de Souza, S.-J. Marrink, G. Maglia, *Nat. Chem.* **2021**, *13*, 1192–1199; b) Y. Qing, H. Bayley, *J. Am. Chem. Soc.* **2021**, *143*, 18181–18187.
- [20] a) X. Yao, N.-N. Song, J. Wang, X. Zhao, M.-Y. Cheng, J. Zuo, K. Qiu, *Front. Chem.* **2021**, *9*, 732378; b) M. Wanunu, W. Morrison, Y. Rabin, A. Y. Grosberg, A. Meller, *Nat. Nanotechnol.* **2010**, *5*, 160–165; c) R. Nouri, Z. Tang, W. Guan, *ACS Sens.* **2019**, *4*, 3007–3013; d) Y. Zhang, G. Wu, W. Si, J. Ma, Z. Yuan, X. Xie, L. Liu, J. Sha, D. Li, Y. Chen, *Nanoscale* **2017**, *9*, 930–939; e) R. C. A. Versloot, F. L. R. Lucas, L. Yakovlieva, M. J. Tadema, Y. Zhang, T. M. Wood, N. I. Martin, S. J. Marrink, M. T. C. Walvoort, G. Maglia, *Nano Lett.* **2022**, *22*, 5357–5364; f) Y. Wang, K. Tian, X. Du, R.-C. Shi, L.-Q. Gu, *Anal. Chem.* **2017**, *89*, 13039–13043.
- [21] a) X. Li, Y.-L. Ying, X.-X. Fu, Y.-J. Wan, Y.-T. Long, *Angew. Chem. Int. Ed.* **2021**, *60*, 24582–24587; *Angew. Chem.* **2021**, *133*, 24787–24792; b) S.-C. Liu, M.-X. Li, M.-Y. Li, Y.-Q. Wang, Y.-L. Ying, Y.-J. Wan, Y.-T. Long, *Faraday Discuss.* **2018**, *210*, 87–99; c) J. Sun, A. K. Thakur, L. Movileanu, *Proteomics* **2022**, *22*, 2100077.
- [22] a) S. Zhang, Z. Cao, P. Fan, Y. Wang, W. Jia, L. Wang, K. Wang, Y. Liu, X. Du, C. Hu, P. Zhang, H.-Y. Chen, S. Huang, *Angew. Chem. Int. Ed.* **2022**, *61*, e202203769; *Angew. Chem.* **2022**, *134*, e202203769; b) H.-J. Hwang, J.-S. Kim, J. Lee, J. S. Min, K.-B. Jeong, E. Kim, M.-K. Lee, S.-W. Chi, *Anal. Chem.* **2022**, *94*, 7449–7454; c) F. Li, M. A. Fahie, K. M. Gilliam, R. Pham, M. Chen, *Nat. Commun.* **2022**, *13*, 3541.
- [23] a) Y. Liu, K. Wang, Y. Wang, L. Wang, S. Yan, X. Du, P. Zhang, H.-Y. Chen, S. Huang, *J. Am. Chem. Soc.* **2022**, *144*, 757–768; b) C. Wloka, V. Van Meervelt, D. van Gelder, N. Danda, N. Jager, C. P. Williams, G. Maglia, *ACS Nano* **2017**, *11*, 4387–4394; c) R. C. A. Versloot, S. A. P. Straathof, G. Stouwie, M. J. Tadema, G. Maglia, *ACS Nano* **2022**, *16*, 7258–7268.
- [24] a) M.-Y. Li, Y.-L. Ying, J. Yu, S.-C. Liu, Y.-Q. Wang, S. Li, Y.-T. Long, *JACS Au* **2021**, *1*, 967–976; b) M.-Z. Huo, M.-Y. Li, Y.-L. Ying, Y.-T. Long, *Anal. Chem.* **2021**, *93*, 11364–11369.

Manuscript received: July 7, 2022

Accepted manuscript online: September 9, 2022

Version of record online: October 5, 2022